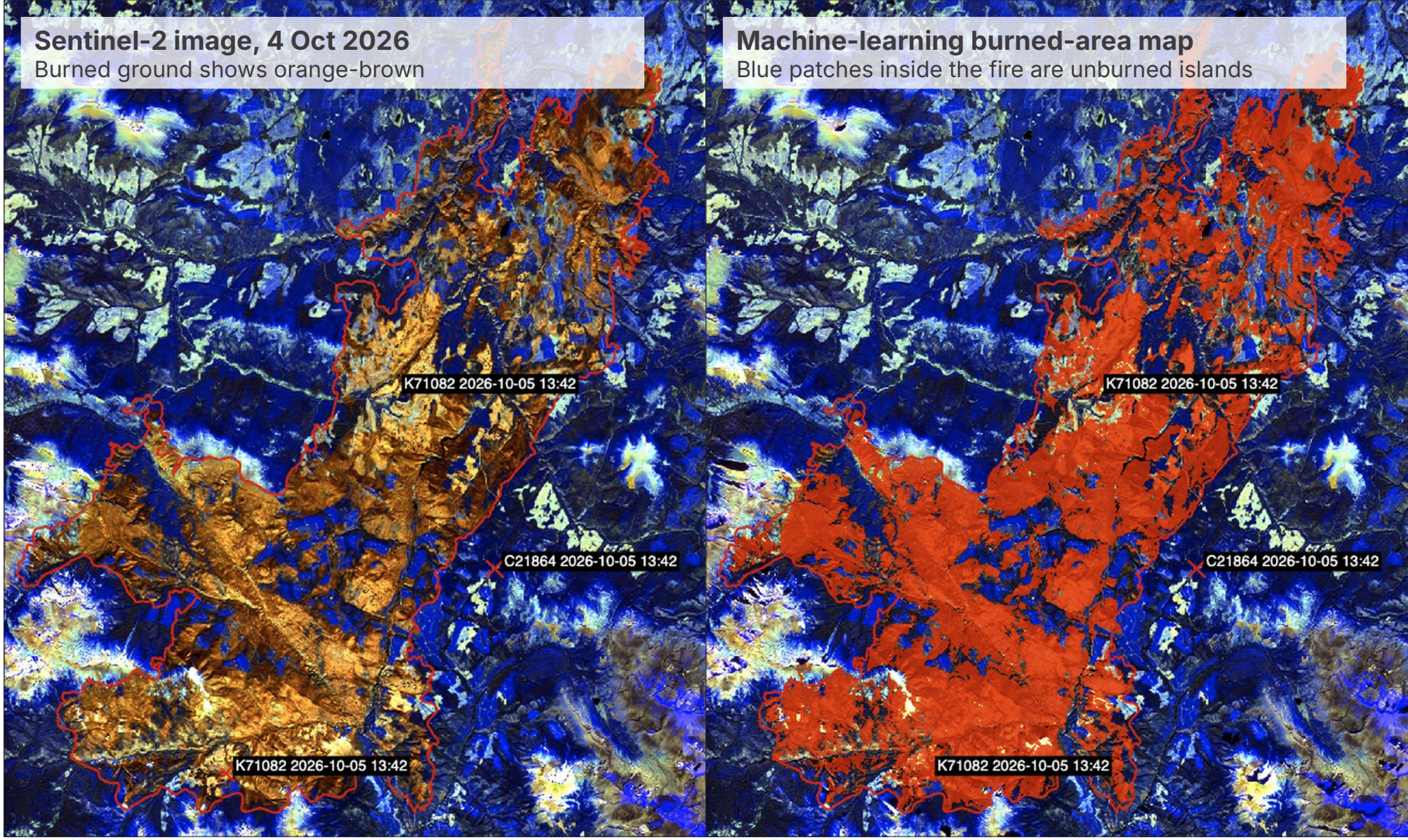
Burned-area mapping with machine learning

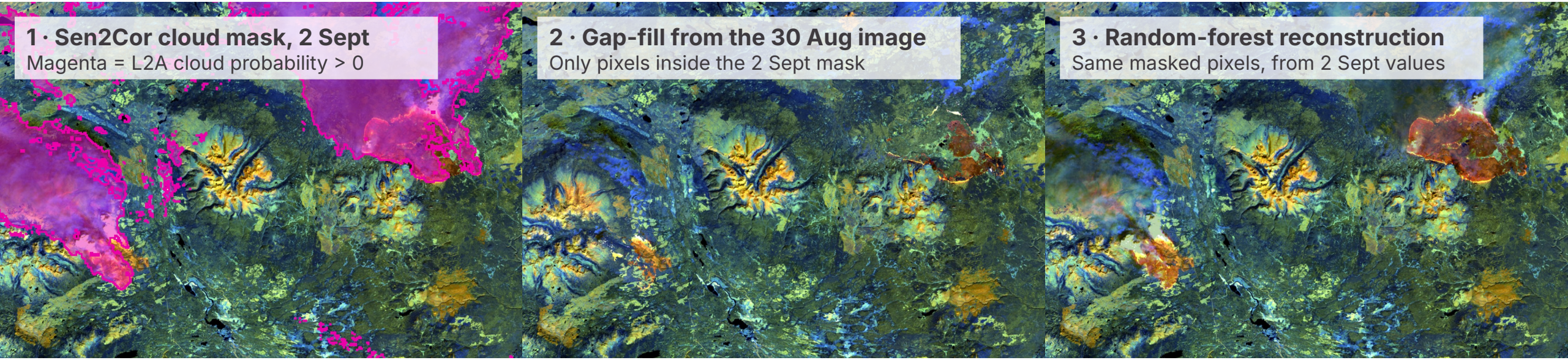
- Clustering groups pixels by spectral signature — no training data needed
- Hotspots or the perimeter pick the burned group
- Analyst reviews, revises and accepts: ML drafts, a person signs off
- Shows unburned islands; refreshes with each clear image



Fire K71082, 2026. Sentinel-2 imagery; perimeter 5 Oct 2026.

Seeing through smoke with machine learning

Fire C11659 · Sentinel-2 tile T09UYU · smoky image 2 Sept 2025 · clear reference image 30 Aug 2025



Smoke is flagged as cloud

Sen2Cor, ESA's standard Sentinel-2 processor, gives the 2 Sept smoke a cloud probability above 0, including over the burning fire edges.

Gap-filling erases the new burn

Each 2 Sept pixel inside the mask takes its 30 Aug value, rescaled per band from pixels clear on both dates. Smoke goes; the fire reverts to its 30 Aug extent.

Random forest keeps new burn

Trained to map smoky 2 Sept values to the step 2 values, it predicts each masked pixel from its own 2 Sept value. Smoke thins; the 2 Sept burn stays.

Proof of concept. In development. Thick smoke over the fire front is still a limit. All pixel values: L1C, native 20 m (RGB = B12, B11, B8A). Cloud masks only: L2A MSK_CLDPRB, 20 m.

2 Sept L1C image S2B_MSIL1C_20250902T192909_N0511_R142_T09UYU_20250902T230923 **30 Aug L1C image** S2A_MSIL1C_20250830T193831_N0511_R142_T09UYU_20250831T003315
2 Sept L2A mask S2B_MSIL2A_20250902T192909_N0511_R142_T09UYU_20250902T231719 **30 Aug L2A mask** S2A_MSIL2A_20250830T193831_N0511_R142_T09UYU_20250831T011715

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Would AI put out a wildfire?